Direct Displacement-Based Design (DDBD)

with Nonlinear Dynamic Time-History Analysis in OpenSeesPy

Complete Code Explanation and All Formulas

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1. Overview

This Python script implements the Direct Displacement-Based Design (DDBD) procedure for a single-degree-of-freedom (SDOF) system and verifies the design by performing a nonlinear dynamic time-history analysis (THA) in OpenSeesPy. The workflow consists of six main stages:

- Definition of target performance parameters (displacement, ductility, equivalent damping).
- Construction of a multilinear force–displacement backbone.
- Evaluation of the design displacement spectrum and solution for the effective period.
- Calculation of the required design strength (base shear) and scaling of the backbone.
- Assembly of an OpenSees model with parallel hysteretic springs and Rayleigh damping.
- Execution of the transient analysis and graphical comparison of results against the DDBD target.

The method follows the principles described by Priestley, Calvi, Kowalsky and subsequent researchers, and is illustrated with a simple bilinear displacement spectrum that can be replaced by any code spectrum (e.g., NZS 1170.5 or Eurocode 8).

2. DDBD Parameters

2.1 Target Performance and Ductility

The design starts from a prescribed target displacement at the effective height of the equivalent SDOF system:

$$\Delta d = 0.015\ m \quad (\text{example corresponding to 2.5 \% drift for } H = 1\ m)$$

An estimate of the yield displacement is also supplied:

$$\Delta y,est = 0.008\ m$$

The displacement ductility demand is then simply:

$$\mu = \Delta d / \Delta y,est$$

2.2 Equivalent Viscous Damping

For reinforced-concrete frames an empirical expression based on the Takeda hysteresis rule is commonly used:

$$\xi eq = 0.05 + 0.565 \cdot (\mu - 1) / \mu$$

The first term represents the inherent (elastic) damping; the second term accounts for the hysteretic energy dissipation that increases with ductility.

2.3 Multilinear Force–Displacement Backbone

A four-point backbone is defined (units: N and m):

Fy, Dy – *yield force and displacement*
$Fu = 1.2\,Fy, Du = 10\,Dy$ – *peak (ultimate) point*
$Fr1 = Fy/4, Dr1 = 12\,Dy$ – *first residual / post-peak point*
$Fr2 = Fy/10, Dr2 = 15\,Dy$ – *second residual point*
These points are stored in two arrays:

$$Force = [Fy, Fu, Fr1, Fr2]$$

$$Displacement = [Dy, Du, Dr1, Dr2]$$

A vector of stiffness scaling coefficients Coff multiplies the backbone of each parallel spring. When all entries equal 1.0 the springs are identical; different values produce a set of springs with proportional strength and stiffness while preserving the same shape of the nonlinear curve.

2.4 Initial Stiffness and Mass

The initial (elastic) stiffness of the unscaled system is:

$$K0 = (\Sigma\, Ci) \cdot Fy / Dy$$

Given a desired initial period T0 (example: 0.5 s), the corresponding mass is obtained from the SDOF period formula:

$$M = K0 \cdot (T0 / (2\pi))^2$$

3. Design Displacement Spectrum

3.1 Damping Modification Factor

A simple EC8 / New-Zealand-style damping reduction factor is applied:

$$\eta = 1.0 \qquad if\ \xi \le 0.05$$

$$\eta = \max(\sqrt{[10 / (5 + 100\,\xi)]}, 0.55) \quad otherwise$$

3.2 5 % Damped Spectrum

A bilinear displacement spectrum is used for illustration (any code spectrum may be substituted):

$$Sd,5\%(T) = (Sd,max / Tc) \cdot T \quad for\ T \le Tc$$

$$Sd,5\%(T) = Sd,max \qquad for\ T > Tc$$

with example values Sd,max = 0.12 m and Tc = 2.5 s.

3.3 Damped Design Spectrum

$$Sd(T, \xi) = \eta(\xi) \cdot Sd,5\%(T)$$

3.4 Effective Period

The effective period Teff is the period at which the damped spectrum equals the target displacement. It is obtained by solving the nonlinear equation:

$$Sd(Teff, \xi eq) = \Delta d$$

numerically (e.g., with scipy.optimize.fsolve).

4. Required Strength from DDBD

Once Teff is known, the effective stiffness of the equivalent SDOF system is:

$$Ke = 4\,\pi^2\,M / Teff^2$$

The design base shear follows directly:

$$Vb = Ke \cdot \Delta d$$

The original backbone is then scaled uniformly so that its yield force matches the required strength:

$$F_DDBD = F \cdot (Vb / Fy)$$

This preserves the shape of the force–displacement curve while providing the correct strength for the design displacement.

5. OpenSees Model Construction

5.1 Geometry and Mass

A one-dimensional model with two nodes is created. Node 1 is fixed; node 2 carries the mass M.

5.2 Parallel Hysteretic Springs

For each coefficient Ci a uniaxialMaterial of type HystereticSM is defined. The positive and negative branches are obtained by scaling the force ordinates of the backbone by Ci while leaving the displacement ordinates unchanged:

$$pos = [Ci{\cdot}Fy, Dy, Ci{\cdot}Fu, Du, Ci{\cdot}Fr1, Dr1, Ci{\cdot}Fr2, Dr2]$$

$$neg = [-Ci{\cdot}Fy, -Dy, \ldots]$$

Each material is assigned to a zeroLength element connecting nodes 1 and 2. The parallel combination of these elements reproduces the desired multilinear backbone with the correct initial stiffness.

5.3 Rayleigh Damping

Mass-proportional damping is introduced:

$$\alpha M = 2\,\xi\,\omega 0\,, \quad \omega 0 = \sqrt{(K0 / M)}$$

so that the damping ratio at the initial frequency equals the design equivalent damping ξeq. Stiffness-proportional terms are set to zero.

5.4 Ground Motion and Analysis

A Path time series is created from the acceleration record (scaled by GM_FACTOR = g · GM_SCALE). A UniformExcitation pattern applies the motion in the single degree of freedom. The transient analysis uses

the Newmark average-acceleration integrator (γ = 0.5, β = 0.25) together with a Newton–Raphson algorithm and a displacement-increment convergence test.

6. Post-Processing and Verification

After the analysis the script extracts:

- Time history of displacement at the free node.
- Corresponding restoring-force history (from reactions).
- Peak absolute displacement, which is compared with the DDBD target Δd.

Four plots are generated:

- Ground-acceleration time history.
- Displacement time history with horizontal lines at ±Δd.
- Force–displacement hysteresis loops.
- Design displacement spectrum with the (Teff, Δd) point marked.

A ratio of achieved-to-target displacement close to unity indicates that the DDBD strength and the chosen ground-motion scale are consistent.

7. Summary of All Formulas

The complete set of governing expressions used in the script is collected below for quick reference.

Quantity	Formula
Ductility	$\mu = \Delta d \,/\, \Delta y,est$
Equivalent damping	$\xi eq = 0.05 + 0.565\cdot(\mu - 1)/\mu$
Initial stiffness	$K0 = (\Sigma\, Ci)\cdot Fy \,/\, Dy$
Mass	$M = K0\cdot(T0/(2\pi))^2$
Damping factor η	$\eta = max(\sqrt{[10/(5+100\xi)]}, 0.55)\ \ (\xi > 0.05)$
5 % spectrum	$Sd,5\%(T) = (Sd,max/Tc)\cdot T\ \ \ (T \le Tc)$
Damped spectrum	$Sd(T,\xi) = \eta(\xi)\cdot Sd,5\%(T)$
Effective period	$Sd(Teff, \xi eq) = \Delta d$
Effective stiffness	$Ke = 4\pi^2\, M \,/\, Teff^2$
Design base shear	$Vb = Ke \cdot \Delta d$
Force scaling	$F_DDBD = F \cdot (Vb \,/\, Fy)$
Rayleigh αM	$\alpha M = 2\,\xi\,\omega 0\,,\ \ \omega 0 = \sqrt{(K0/M)}$

8. References

Sullivan, T.J. – Highlighting Differences between Force-Based and Displacement-Based Design Solutions for Reinforced Concrete Frame Structures.

30 Years of Direct Displacement-Based Design: Where We Started, What's Been Done, Where We're Going (WCEE 2024).

Priestley, M.J.N., Calvi, G.M., Kowalsky, M.J. – Displacement-Based Seismic Design of Structures.

OpenSees / OpenSeesPy documentation: HystereticSM material, zeroLength element, Rayleigh damping, Newmark integrator, UniformExcitation.